



Questions on E/R Theory

1.
ER was developed by Chen in 1980.
 - A. True
 - B. False
2.
ER was developed by Chen in 1976.
 - A. True
 - B. False
3.
ER aims to:
 - A. Facilitate database programming
 - B. Facilitate database organization
 - C. Facilitate database design
4.
The Conceptual data model is the set of concepts that:
 - A. Describe the structure of a database and the associated insert and update transactions
 - B. Describe the structure of a database and the associated retrieval and update transactions
 - C. Describe the structure of a database and the associated retrieval transactions
5.
An ERs purpose is to support a user's perception of the data and conceal the technical aspects associated with database design.
 - A. True
 - B. False
6.
A high-level conceptual model is dependent on the DBMS and HW platform used to implement the database.
 - A. True
 - B. False
7.
The basic concepts of the Entity-Relationship model include:
 - A. Entity types
 - B. File types
 - C. Relationship types
 - D. Attributes
 - E. Storage details



8.
Entity type = An object or concept identified by the enterprise as having an _____ existence.
9.
A Entity is
- A. An instance of an Entity type that is uniquely identifiable
 - B. An instance of a DBA type that is uniquely identifiable
 - C. An instance of a type that is uniquely identifiable
10.
Weak entity type:
- A. An entity that is not existence-dependent on some other entity type
 - B. An Entity that characterizes a DBMS
 - C. An instance that is existence dependent on some other entity type
11.
Strong entity type:
- A. An entity that is not existence-dependent on some other entity type
 - B. An Entity that characterizes a DBMS
 - C. An instance that is existence dependent on some other entity type
12.
An Attribute is a property of an entity or a relationship type.
- A. True
 - B. False
13.
An Attribute:
- A. a set of values that might be assigned to an attribute
 - B. is a property of an entity or a relationship type
14.
An Attribute domain:
- A. a set of values that might be assigned to an attribute
 - B. is a property of an entity or a relationship type
15.
Simple attribute:
- A. An attribute that represents a value that is derivable from a related attribute or set of attributes, not necessarily in the same entity
 - B. An attribute that holds multiple values for a single component
 - C. An attribute that holds a single value for a single entity
 - D. An attribute composed of multiple components each with individual existence
 - E. An attribute composed of a single component with an independent existence



16.

Composite attribute:

- A. An attribute that represents a value that is derivable from a related attribute or set of attributes, not necessarily in the same entity
- B. An attribute that holds multiple values for a single component
- C. An attribute that holds a single value for a single entity
- D. An attribute composed of multiple components each with individual existence
- E. An attribute composed of a single component with an independent existence

17.

Single valued attribute:

- A. An attribute that represents a value that is derivable from a related attribute or set of attributes, not necessarily in the same entity
- B. An attribute that holds multiple values for a single component
- C. An attribute that holds a single value for a single entity
- D. An attribute composed of multiple components each with individual existence
- E. An attribute composed of a single component with an independent existence

18.

Multi- valued attribute:

- A. An attribute that represents a value that is derivable from a related attribute or set of attributes, not necessarily in the same entity
- B. An attribute that holds multiple values for a single component
- C. An attribute that holds a single value for a single entity
- D. An attribute composed of multiple components each with individual existence
- E. An attribute composed of a single component with an independent existence

19.

Derived attribute

- A. An attribute that represents a value that is derivable from a related attribute or set of attributes, not necessarily in the same entity
- B. An attribute that holds multiple values for a single component
- C. An attribute that holds a single value for a single entity
- D. An attribute composed of multiple components each with individual existence
- E. An attribute composed of a single component with an independent existence

20.

A Candidate key is an attribute or set of attributes that uniquely identifies individual occurrences of an entity type.

- A. True
- B. False

21.

The Primary key:

- A. Uniquely identifies each occurrence of an entity type
- B. Is equivalent with a composite key
- C. Is the candidate key selected to be a primary key



22.

A Composite key is a candidate key that consists of two or more attributes.

- A. True
- B. False

23.

Degree of a relationship.

- A. Is always 1
- B. The number of participating entities in a relationship
- C. The number of representations in a relationship